

When to Challenge a Pitch? A Reinforcement Learning Approach to ABS Challenge Strategy

"Why the Best Move is Often to Hold"

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“The Limited Resource” When should a player challenge a pitch?

<https://www.mlb.com/video/edwin-diaz-ball-to-randy-arozarena-we8lw>

Successful Challenge

MLB All-Star Game (Top 9th): With an 0-2 count and two outs, Edwin Díaz challenges a called ball. The call is overturned to strike three, ending the inning.

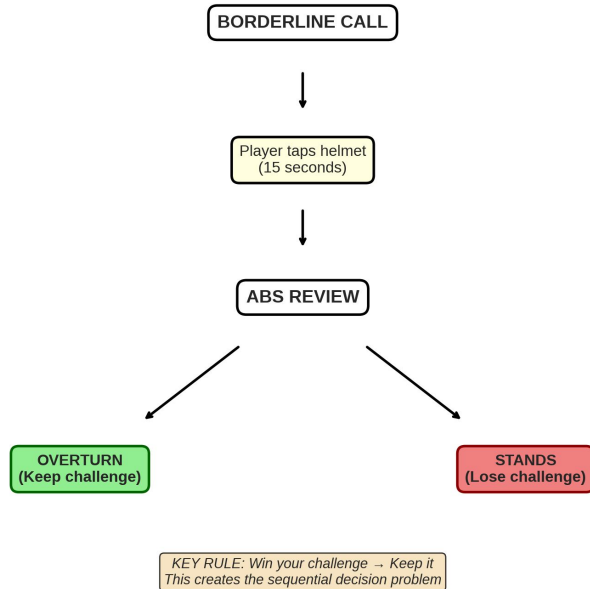
<https://www.mlb.com/video/giants-challenge-pitch-strike-is-confirmed-for-k>

Unsuccessful Challenge

Giants vs. Dodgers (3rd inning): With a 2-2 count, Jerar Encarnacion challenges a called strike with one out. The call is confirmed, resulting in a strikeout for the second out of the inning.

MLB's 2026 ABS Challenge System

ABS Challenge Flow



Key Facts:

- Limited resource: 2 challenges per team per game
- Successful challenge is retained
- Only the batter, catcher, or pitcher can initiate a challenge
- The challenge request must be immediate (player-initiated, in the moment)
- No coach input on the decision
- Triple-A testing: teams averaged about ~4 challenges per game with about ~50% overturn rate

“Greedy vs. Patient” Two philosophies

GREEDY

- Challenge if: $P(\text{overturn}) \times \Delta\text{Value} > 0$
- Challenge when EV is positive NOW
- Traditional approaches: Challenge immediately when probability of winning a call looks good

Patient

- Consider: What's this challenge worth later?
- Our approach: Treat as sequential decision-making problem
- Maximize long term-expected value (Saving this call for later is worth more than the risk of losing it now plus the gain of winning the current call)
- Consider future value of saved challenges

Which approach wins more games?

Previous Work: One-Step Optimization

Challenge if: $P(\text{overturn}) \times \Delta\text{WE} > \text{threshold}$

"Greedy" One Step optimization

Challenge if: $P(\text{overturn}) \times \Delta\text{WE} - \lambda(1 - P(\text{overturn})) > 0$

- Uses XGBoost to estimate $P(\text{overturn})$
- Calculates immediate win expectancy change
- Adds penalty λ for failed challenges

Limitation: Does not consider future opportunities
Misses sequential nature of the problem: option value of saving challenges

Umpires miss ~10% of borderline calls (Triton Ball 2025)

The Missing Piece: Option Value

	Single-Step	Sequential
Simple P(overtake)	Historical (gut feel)	-
Sophisticated P(overtake)	Triton Ball (XGBoost)	This research

First application of reinforcement learning to ABS challenge strategy

The limitations (the important part)

Greedy policies miss multi-step consequences, especially under your exact rules:

- Option value of holding: If you challenge now, you might lose a challenge and be unable to challenge a later, higher-leverage pitch.
- Retention creates path dependence: Successful challenges effectively preserve future flexibility, which greedy one-step scoring does not capture cleanly.
- Game context evolves: Base-out state, inning, score differential, and count evolve. A short-term gain can be dominated by saving the resource for later states.

Framing the problem: Markov Decision Process (How Q -Learning Works)

STATE: "What's happening?"

ACTION:
"Challenge or wait?"

REWARD:
"Expected runs gained"

Components:

- State: Game context (inning, outs, count, runners, score, challenges left)
- Action: Challenge or Hold
- Reward: Change in run expectancy (RE24) if successful
- **Goal: Maximize cumulative expected runs over the season**

State space - "What the Agent Sees" 2,592 discrete states

STATE = (
 Inning: [4 buckets] Early (1-4), Mid (5-7), Late (8-9), Extras (10+)
 Outs: [3 values] 0, 1, 2
 Count: [4 buckets] Hitter ahead, Even, Pitcher ahead, Two-strike
 Runners: [3 buckets] None, Scoring position, Runner on 1st only
 Score: [3 buckets] Behind, Tied, Ahead
 Challenges: [3 values] 0, 1, 2
 Pitch distance: [2 buckets] Close to edge, Not close
)

Total States: $4 \times 3 \times 4 \times 3 \times 3 \times 3 \times 2 = 2,592$

The Reward Function: "How We Keep Score" Run Expectancy (RE24)

RE24 Matrix

Base State	0 Outs	1 Out	2 Out
Empty	0.481	0.254	0.098
1st	0.859	0.509	0.224
2nd	1.100	0.664	0.319
1st & 2nd	1.437	0.884	0.429
3rd	1.350	0.950	0.353
1st & 3rd	1.784	1.130	0.478
2nd & 3rd	1.920	1.352	0.570
Bases Loaded	2.254	1.546	0.752

- RE24 = expected runs from each base-out state
- Successful challenge improves your RE24 position
- Failed challenge = 0 reward but lose a challenge

Reward = RE24(after) - RE24(before) if overturn,
else 0

Q-learning algorithm: "Teaching the Agent" 100 epochs of simulated baseball

The Bellman update equation:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

Parameter	Value	Meaning
α (learning rate)	0.1	How fast to update
γ (discount)	0.9	Future value weight
ϵ (exploration)	0.5 $\rightarrow$ 0.02	Explore vs exploit
Epochs	100	Training iterations
Games	1,097	Training games

Takeaways

- Standard Q-learning with Dyna updates
- Exploration decays from 50% to 2%
- Trained on 1,097 games (64,954 pitches)

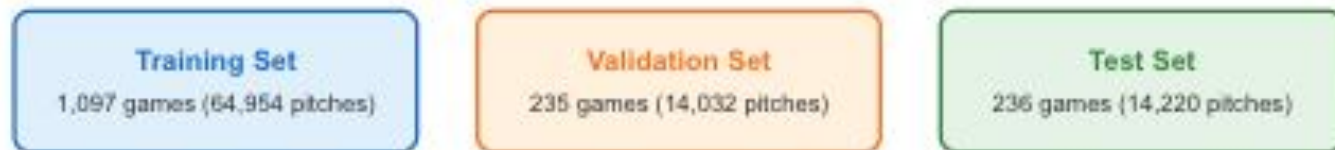
Agent learns: $Q(\text{state}, \text{action}) = \text{immediate reward} + \text{expected future rewards}$

Data Pipeline and The Dataset: 93,206 pitches from 2024

Data Pipeline: 93,206 Pitches → Q-Learning Policy



93,206 pitches → 1,568 games



Split by GAME (not pitch) to prevent data leakage

→ Q-Learning Training →

Data & Methodology Notes

- Data: MLB Statcast (June-Sept 2024) simulating ABS challenge outcomes
- Challenge outcomes modeled from pitch location distance to zone edge
- Train/test split by GAME (prevents data leakage)
- 93,206 pitches from 1,568 games

Policy Comparisons: Terminal-Count vs. Greedy vs. Q-Learning

	Terminal-count	Greedy (Triton Ball-style analytics)	Q-Learning (sequential optimization)
Decision Rule	Heuristics (gut feel) only on 4 ball counts or 3 strike counts	$P(\text{overturn}) \times \Delta \text{RE24} > 0$	$Q(s, \text{CHAL}) > Q(s, \text{HOLD})$
Future Consideration	No	No	Yes
Basis	Manager intuition	Immediate EV	Learned value function
Represents	Status quo, “common sense - challenge on the pitche that results in the out”	Smart analytics (Triton Ball-style analytics)	This research with reinforcement learning

Results: Q-Learning Performance

Policy	ER/Game	Success Rate	Challenges/ Game	Wins/Season
Terminal-Count	0.0438	24.1%	2.64	+0.71
Greedy	0.0414	22.2%	2.57	+0.67
Q-Learning	0.1164	28.9%	2.78	+1.89

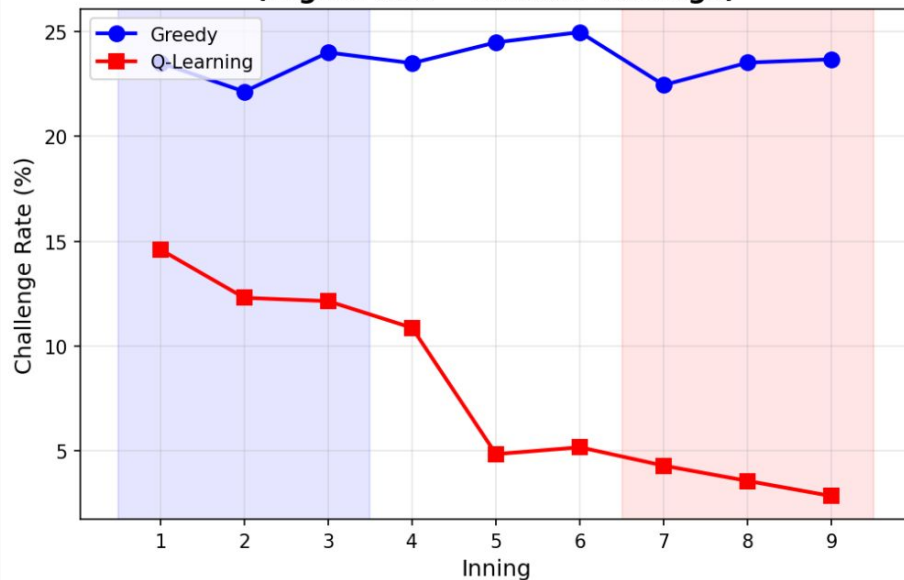
Takeaways:

- Q-Learning: 0.1164 ER/game vs 0.0414 for Greedy and 0.0438 for Terminal-Count
- All three challenge ~2.6-2.8 times per game
- The difference is WHEN, not HOW MUCH

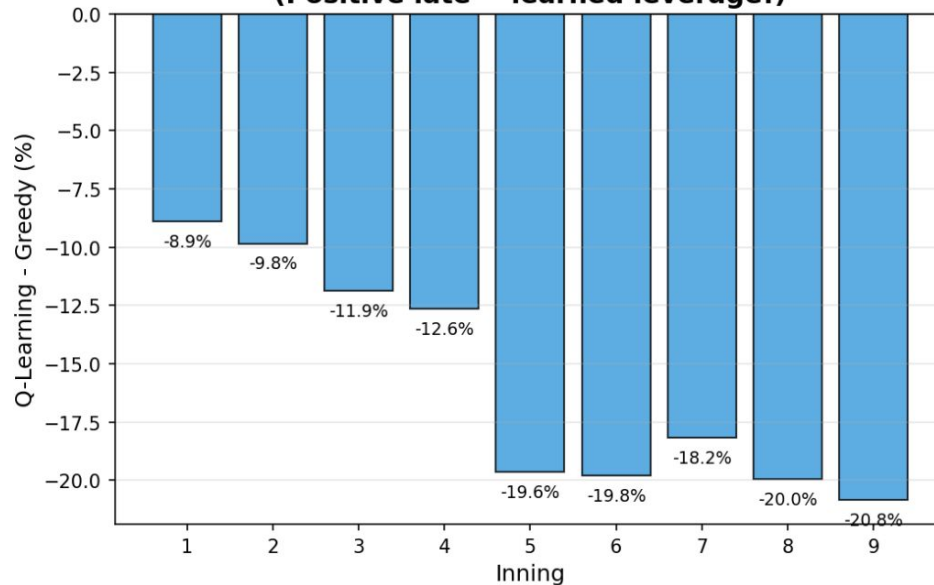
Q-Learning Learns to Be Selective

Q-Learning is 2.8× More Effective Per Game but tends to be more selective

Challenge Rate by Inning
(Higher late = learned leverage)



Q-Learning Challenge Rate Difference
(Positive late = learned leverage!)

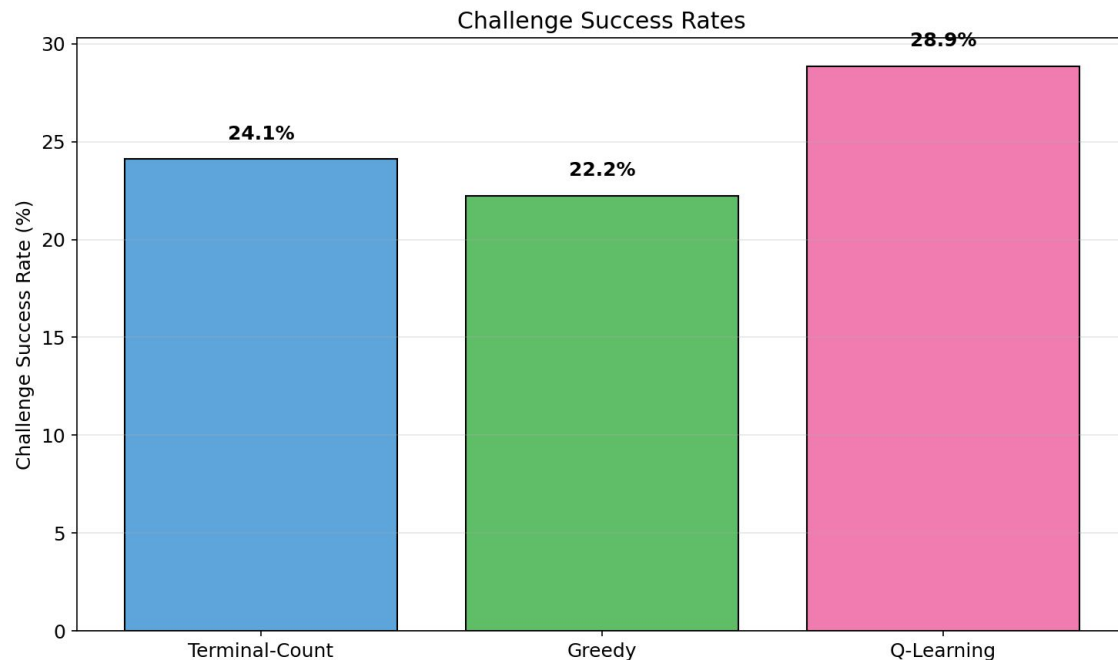


Challenge Distribution by Inning

Inning	Greedy	Q-Learning	Diff
1-4 avg	23.3%	12.5%	-10.8%
7-9 avg	23.2%	3.6%	-19.6%

- Q-Learning is Highly Selective Throughout the Game
- This is learned behavior, not programmed

Better Timing = Higher Success Rate

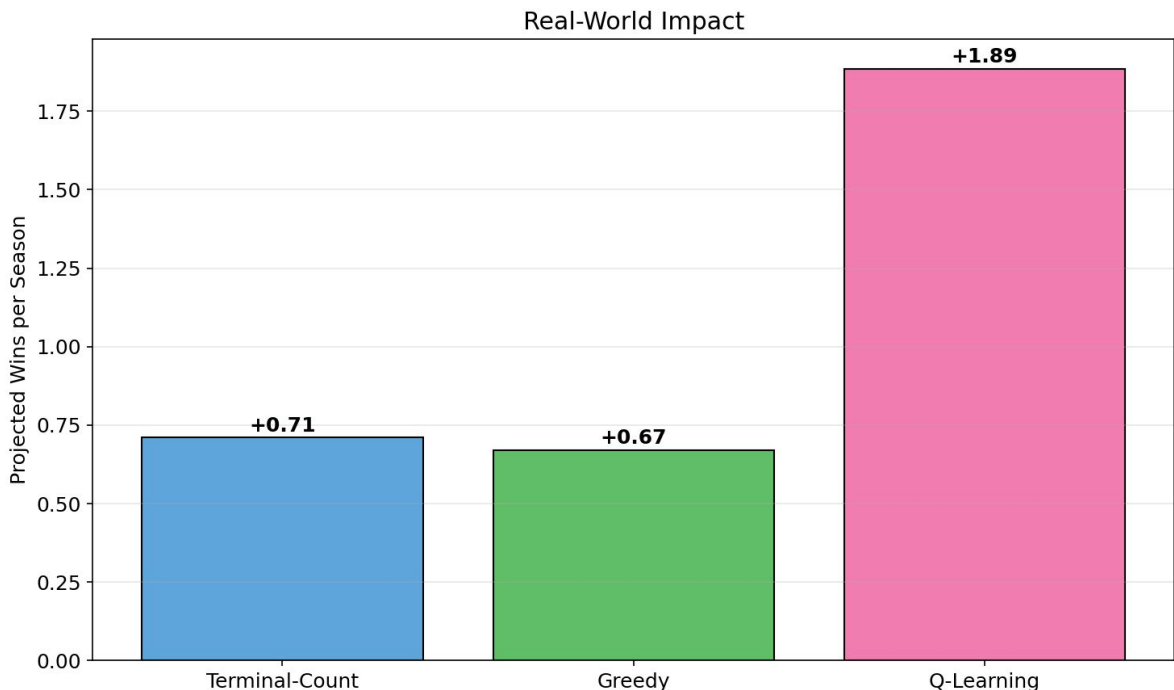


Insight: Better timing = Higher overturn rate

- Q-Learning has higher overturn rate
- Not just about challenging more, but challenging smarter
- Being selective means higher success
- a graph shows the success rate

Q-Learning wins 6.7 percentage points more challenges

Real-World Impact: +1.22 Wins Over Greedy Baseline



Context:

- In 2024, 6 teams within 2 games of playoffs - every half-win matters

Takeaways:

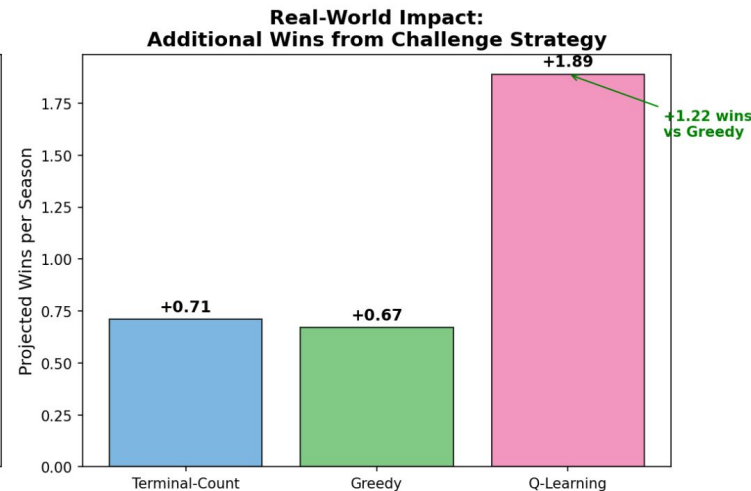
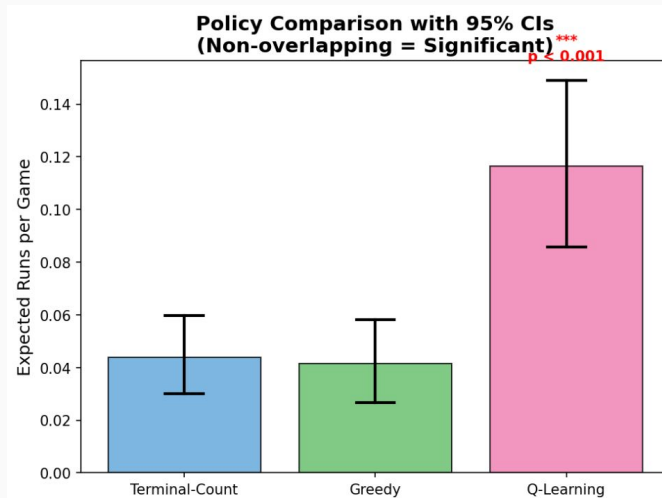
- Q-Learning absolute: +1.89 wins/season (vs never challenging)
- Q-Learning improvement over Greedy: +1.22 wins/season
- Q-Learning improvement over Terminal-Count: +1.18 wins/season

Statistical Significance

Key Takeaways:

- $p < 0.001$, meaning less than 0.1% chance this is random
- Cohen's d: 0.38 (small-to-medium effect)
- 95% CI: [0.086, 0.149]

- 95% Confidence Interval: [0.086, 0.149] ER/game, which translates to approximately [+1.4, +2.4] wins/season



How Did We Validate This?

- Train/Test Split: 1,097 games (64K train, 14K test)
- Bootstrap Resampling: 10,000 iterations
- 95% Confidence Interval: [0.086, 0.149] ER/game, which translates to approximately [+1.4, +2.4] wins/season
- Non-overlapping CIs = Statistically significant

Why Q-Learning Wins: Three Behaviors the Agent Learned (Not Programmed)

Option Value: The worth of keeping a resource available for future use, even if using it now has positive expected value.

Three Learned Behaviors:

1. **Selectivity:** Only challenges high-confidence opportunities (28.9% success vs 22.2%)
2. **Resource** Management: Understands cost of using vs. saving
3. **Patience:** Waits for truly valuable opportunities

Takeaways

- Holding has value (option value)
- The agent learned THREE key behaviors
- None of these were programmed - they emerged from training
-

Example game scenario: How Policies Differ

6th Inning

Situation: Tied 2-2, runner on 2nd, 1 out

Pitch: Borderline strike,
 $P(\text{overtake}) = 65\%$

ΔRE_{24} : 0.04 runs

GREEDY: Challenge! ($0.65 \times 0.04 = +0.026 > 0$) ✓

Q-LEARNING: Hold. ($Q_{\text{hold}} > Q_{\text{challenge}}$) ✗

9th Inning

Situation: Tied 2-2, bases loaded, 2 outs

Pitch: Same borderline strike,
 $P(\text{overtake}) = 65\%$

ΔRE_{24} : ΔRE_{24} : 0.15 runs

GREEDY: No challenges left!

Q-LEARNING: Q-LEARNING:
Challenge! (High-value situation) ✓

Key Takeaways:

- Greedy uses challenge on marginal 6th-inning opportunity
- Q-Learning saves it for high-leverage 9th-inning moment
- Same pitch, different context, different decision
- note Q-Learning selectivity, not just timing

Example of Wasted Challenge

<https://www.mlb.com/yankees/video/rays-challenge-pitch-ball-four-call-is-confirmed?>

Unsuccessful Challenge (Wasted Challenge Example)

Rays vs. Yankees: A ball four call is challenged, but the ABS review confirms the original call. This is a “wasted” challenge because it spends a limited resource in a spot with a low chance of overturn, creating opportunity cost later in the game.

Comparison to Triton Ball's Approach

Dimension	Triton Ball (2025)	This Research
Approach	XGBoost for P(overturn)	Q-Learning for WHEN
Optimization	One-step greedy	Multi-step sequential
Result	Better call prediction	Better timing strategy

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Limitations & Future Work - What I Didn't Model

Current Limitations:

- No pitcher-specific tendencies
- No umpire-specific probability adjustments
- Trained on Triple-A proxy data
- Simplified state space

Future Research Directions:

- Integrate umpire zone accuracy (combine with Triton Ball's XGBoost)
- Add pitcher command models
- Validate on real 2026 ABS data
- Deep RL for richer state representation
- Team-specific policy customization

Main Findings

Main Findings:

1. Sequential decision-making matters: Q-Learning outperforms Greedy by +1.22 wins/season
2. Selectivity wins: Q-Learning achieves 28.9% on challenge success rate vs Greedy's 22.2%
3. Framework ready for 2026 deployment

For Front Office: Implementing the Strategy

What Q-Learning Discovered:

THE KEY INSIGHT: Be selective. Very selective.

Early Innings (1-6):

- Challenge rate: ~12% (vs Greedy's 23%)
- Only challenge OBVIOUS misses
- "If you're not sure, don't challenge"

Late Innings (7-9):

- Challenge rate: ~4% (vs Greedy's 23%)
- Be EVEN MORE selective
- Having challenges available IS valuable

THE RESULT:

- Q-Learning wins 28.9% of challenges
- Greedy wins only 22.2%
- Same number of total challenges, better outcomes

Ultimate result: +1.22 wins/season from better selectivity

- **The Counterintuitive Late-Inning Rule**
- **The "Successful Challenge Trap"**

Final Thought

"The Edge That Wins Divisions"
In a sport where 1-2 wins
separate playoff teams from
eliminated teams, optimal
challenge strategy isn't just
academic - it's the kind of edge
that wins divisions.

AL LEADERS		W	L	PCT
	Toronto Blue Jays - E z	94	68	.580
	Seattle Mariners - W y	90	72	.556
	Cleveland Guardians - C y	88	74	.543
AL WILD CARD		W	L	PCT
	New York Yankees w	94	68	.580
	Boston Red Sox w	89	73	.549
	Houston Astros	87	75	.537
	Detroit Tigers w	87	75	.537
	Kansas City Royals	82	80	.506

NL WILD CARD		W	L	PCT	WCGB
	Chicago Cubs w	92	70	.568	+9.0
	San Diego Padres w	90	72	.556	+7.0
	New York Mets	83	79	.512	-
	Cincinnati Reds w	83	79	.512	-

Thank You!

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